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REAL-TIME NOISE FILTERING WITH ADAPTIVE FILTERS IN HEAVY EQUIPMENT SOUNDSCAPE

Master of Science Thesis
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ABSTRACT

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In this master's thesis, adaptive filters are used to abate the engine noise of heavy equipment and the changes in the soundscape are studied. The main objective of this work is to enhance the sound quality of human speech and shouting. The results are evaluated both with subjective tests and computationally. The first method used in achieving the goal is the Butterworth bandpass-filter, which is designed to preserve the frequencies possibly containing human speech and filter the rest of the frequencies. The resulting signal of the Butterworth filter is filtered with the adaptive filter targeting the engine noise. In this study two different adaptive filters are used, the Wiener filter and NLMS filter, and their performance is compared. In addition to the method of implementation, these filters also differ in that the Wiener filter does not use a reference signal for adaptation, but the noise is estimated from the input signal itself, while the NLMS filter uses a microphone in the engine compartment as a reference signal. The filtering system developed in this study is implemented on a developing platform, which is designed to be used by the end user, in other words, the operator of the heavy equipment. This is the reason why the results of the subjective tests are in focus in this study. According to both objective and subjective evaluations in this study, the engine noise deteriorates considerably and the speech coming outside the vehicle is cleaner. In the thesis the results were evaluated both with Sandvik Pantera DPI series drilling machine and a large diesel engine car, Nissan Pathfinder. The subjective evaluation is compared with the signal-to-noise ratio and signal-distortion ratio, which both indicated that the enhancement was successful. Even though the test conditions were not optimal, the results show that the adaptive filters can be used efficiently to filter the engine noise in real-time.

Keywords: Wiener filter, NLMS, bandpass filter, engine noise filtering, adaptive filtering

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Tässä työssä käytetään adaptiivisia suotimia työkoneen moottoriäänen reaaliaikaiseen suodattamiseen ympäristöäänestä ja tutkitaan muuttuuko äänimaailma paremmaksi. Pääasiallisena tavoitteena työssä on ihmisen puheen ja huudon äänenlaadun parantuminen. Tätä mitataan työssä sekä subjektiivisessa testissä että laskennallisesti. Menetelmänä suodatuksessa on käytetty ensin Butterworth-kaistanpäästösuodatinta, joka on säädetty päästämään läpi ne taajuudet, joilla ihmisäänet todennäköisimmin kuuluvat. Tämän jälkeen signaali vielä suodatetaan meluun adaptoituvalla suotimella. Tällaisia suotimia työssä ovat Wiener- ja NLMS-suotimet, joiden toimintaa vertaillaan keskenään. Toteutustavan lisäksi suotimet eroavat toisistaan myös siten, että Wiener-suodin ei käytä referenssiääntä adaptoitumiseen, vaan melu estimoidaan itse signaalista, kun taas NLMS-suotimen kanssa käytetään moottoritilassa olevaa mikrofonia referenssisignaalina. Työssä kehitetty suodinjärjestelmä on implementoitu kehitysalustalle, jonka on tarkoitus toimia loppukäyttäjän, toisin sanoen työkoneen kuljettajan käytössä ja olla tälle mahdollisimman miellyttävä kuunnella. Diplomityötä varten tehdyn subjektiivisen evaluoinnin mukaan moottorin ääni vaimenee huomattavasti ja ajoneuvon ulkopuolelta tuleva puhe selkeytyy. Diplomityössä suodatustulosta arvioitiin sekä Sandvikin Pantera DPI sarjan porakoneella, että dieselmoottorisella Nissan Pathfinder -autolla. Subjektiivisen evaluoinnin tulosta verrataan työssä signaali-kohinasuhteeseen (signal-to-noise ratio) ja signaali-vääristymäsuhteeseen (signal-distortion ratio), jotka indikoivat myös melun vaimennuksen onnistuneen. Vaikka testiolosuhteet eivät olleetkaan optimaalisia, tulokset osoittavat adaptiivisten suotimien sopivan hyvin moottorimelun suodattamiseen.

Avainsanat: Wiener suodin, NLMS, kaistanpäästösuodin, moottoriäänen suodatus, adaptiivinen suodatus

Tämän julkaisun alkuperäisyys on tarkastettu Turnitin OriginalityCheck -ohjelmalla.

PREFACE

This is the masters of science thesis in the computing sciences, specializing in signal processing, in Tampere University. The thesis is based on my work in a signal processing company Meluta Oy, more closely in a project called Navigation Innovation and Support Programme (NAVISP) – Sound Monitor for Autonomous Machines (SMAM) organized by European Space Agency (ESA). I found this wonderful workplace with the help of professor Tuomas Virtanen, so thank you for the hint. I would also like to thank the examiners of this thesis, Tuomas Virtanen and Okko Räsänen.

Huge thanks to Meluta for providing me the best place I have worked so far. The biggest thanks however go to the people in Meluta. Specially the advisor of this thesis and the head of Meluta Markku Salmela, who always stayed positive and steered the NAVISP SMAM project forwards. For the technical guidance I give my thanks to Kari Järvinen and Olli Ali-Yrkkö, without you I would have lost my mind with trying to get over all the difficulties and bugs in my code. For Henri Laakso, thank you for helping me with the listening tests and acting as a peer support with the writing of the thesis.

Last but not least I would like to say thanks to my family. My husband Erno Vanhala read my thesis through many times and helped me to tackle the formatting difficulties and spelling errors. Also, thanks to my son Felix, who provided me many wonderful possibilities to disengage from this work.

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Aino Koskimies

CONTENTS

LIST OF SYMBOLS AND ABBREVIATIONS

ANC	Active Noise Control
β	normalized step-size
CMOS	Comparative Mean Opinion Score
e, ε	error vector
η	local neighbourhood
fs	Sampling Frequency
H	transfer function
h	filter
IIR	Infinite Impulse Response
j	imaginary unit
k	k^{th} element of a vector
λ	eigenvalue of the auto-correlation matrix
M	the length of a vector
m	filter order
MSE	Mean-Square Error
μ	convergence rate, time-varying step-size
N	size of the local neighbourhood
n	noise signal
NHTSA	National Highway Traffic Safety Administration
NLMS	Normalized Least Mean Square filter
ω_c	cut-off frequency
P	power spectrum
p	an estimation of the cross correlation vector
R	an estimation of the autocorrelation matrix
s	sample
SDR	Signal-to-Distortion Ratio
σ^2	variance

SNR	Signal-to-Noise Ratio
SPL	Sound Pressure Level
STFT	Short-Time Fourier Transform
t	time
UI	User Interface
w	filter weights
X	Fourier transform of $x(t)$
x	signal of interest, signal without noise
$\hat{x}$	estimated noiseless signal
Y	Fourier transform of $y(t)$
y	unfiltered signal, noise corrupted signal
z	a complex number

1. INTRODUCTION

When developing autonomous and intelligent machines, the visual sensor technology is the most commonly used and studied technique [1, 2]. However, the usage of other methods, such as audio data, may gain a more comprehensive general view of the surrounding environment. People and other living beings get a wide spectrum of information about their surroundings through many different senses. Taken this into the account there is a need to develop more senses, such as hearing, also to autonomous vehicles. This way they would be developed towards a vehicle that is fully independent from a human driver.

Sometimes senses can also complete each other. For example, the driver can sometimes first hear an emergency vehicle and after that they notice it visually and can decide how to give a way to the emergency vehicle. The same thing happens if the driver sees a bump on the road: while running over it, they can hear or physically feel if the bump was soft or hard and if it has caused any scratches to the car. Therefore, researching the topic of multiple different drive-assisting sensors would possibly turn out to be beneficial for the development of self-driving vehicles.

In order to interact with people in the everyday environment, it is important that autonomous vehicles have sound sensing capability. Big industrial equipment have usually wider blindsides than passenger cars, so the implementation of audio processing into the industrial machines would be even more important than for passenger cars [3]. Sound improves confidence and adds robustness to the visual detection, for example with sound the system can differentiate a stationary human from a same-sized rock – if it is talking it is most probably a human. Improving automation with audio processing capabilities can also help to ease the driving experience and better the accessibility of autonomous vehicles for example for people with limited vision [4].

According to National Highway Traffic Safety Administration (NHTSA) there are six levels of automation starting from no automation to full automation. These serve as a guideline in vehicle automation and are widely used in developing autonomous vehicles. At the highest level (full automation) the car has the control unconditionally and at the lowest level the car only provides momentary assistance for the driver [5]. In this study the operation is done in the levels two (partial automation) and three (conditional automation).

In this study the focus is on improving the autonomous or remotely operated heavy equipment, in this case a drilling machine. The background of this study lays in the need of making Sandvik Pantera DPI series drilling equipment safer by filtering out the engine noise and bringing the remaining environment sounds to the system, remote operator or the driver. The customer of this project, Sandvik Mining and Construction oy, wished that the developed system would specifically help the machine to recognize human presence outside the vehicle by using sound. This requires real-time noise filtering, since the engine noise is so loud that the human sounds are covered by it.

The main reason for focusing on human sounds comes from the safety aspect. On the drilling site, there is not so much other traffic than the drilling equipment and the vehicle's cabin is blocking almost all sounds coming outside the vehicle and the sound of the drill masks the rest of them. By bringing the enhanced soundscape, where the engine and the drill noises are filtered out, to the driver the safety of the surroundings of the drill can be improved.

In the filter solution, adaptive filters such as the Wiener filter and the normalized least-mean square (NLMS) were the most desirable filters. The aforementioned algorithms adapt their filtering parameters to match the given reference noise. Since in this case we have the engine noise available at all times and can be assumed stationary, it can be used as a reference noise, and therefore it is hypothetically possible to get better filtering results than without the reference noise [6]. Besides the adaptive filtering, as the filtering was focused to enhance human speech, the frequencies where there are known to be no human voices or where the engine noise was really loud, but the human voice frequencies are not relevant to get the audible speech were filtered out with a bandpass filter. Since the algorithm is task-specific, there were no off-the-shelf solutions suitable for this problem. In addition, even though many studies on sound enhancement can be found, no similar research topic did not come across as what this thesis represents.

The goal of this study if it is possible to reduce engine noise and enhance speech by using adaptive filters. Also, it is intended to analyze which factors may affect the results of the filtering and how could the challenges be overcome. This goal can be summarized in these research questions: Do the implemented filtering solutions manage to better the original audio? What are the main challenges in real-time processing with adaptive filters considering soundscape and speech enhancement? Does the subjective evaluation differ from the mathematical evaluation?

This thesis is structured as follows. The chapter 2 discusses briefly the basics of active noise cancellation and introduces thoroughly the background of a bandpass filter and the adaptive filters (Wiener filter and normalized LMS filter) regarding the final filtering solution. After that, the speech enhancement in the real-time environment is discussed more comprehensively and the selection of the validation method is justified in relation to the filtering solution. Finally, other methods found in the literature are briefly mentioned.

The chapter 3 summarizes the implementation itself and presents the used platform and algorithms. It also shows figures of the tools and microphone placements. 4 discusses the objective and subjective evaluation results and compares the differences between those two evaluation methods. Also, the comments from the test subjects and from customer's side are summarized and the steps for future development are expounded upon. Finally the chapter 5 summarizes the whole work and discusses the significance of the results of this thesis.

2. THEORETICAL BACKGROUND

This chapter describes the background of the methods used in this research. In the first section the concept of active noise cancellation is explained. The next section tells the background of all used filtering techniques and then the specific characteristics of speech enhancement are elucidated. After that, the background of the objective and subjective validation methods are introduced. Finally, the other methods possibly suitable for this task are considered and the validation methods are explicated.

2.1 Active Noise Control

In audio processing, the term *noise* refers to an audio signal which does not contain any relevant information or is otherwise unwanted by the listener [7]. Noise can also be defined as acoustic energy which is psychologically and physically harmful for people [8]. There are many sources where the noise can come from: acoustic, electromagnetic and electrostatic [9]. In this study, only the acoustic noise is discussed since it is the most common type of noise in speech signals [10]. When this kind of noise is wanted to be removed or reduced, the concept of noise cancellation provides an answer.

There are two types of noise control, active and passive. In passive noise control physical structures, like sound isolating or absorbing building materials, structures, and extensive solutions for vibration controlling, are used to control noise. In addition to difficulties with low-frequency noise management, passive noise cancellation can become immoderately expensive. Active noise control however provides a low-cost solution to this problem. [11]

The active noise control enables the noise dampening to be controlled in a real-time environment irrespective of the traffic and road conditions [12]. It is often used to describe anti-noise techniques, where the power source, most often a wave that has an opposite phase than the noise but the same amplitude, is used to reduce the noise [13]. The input from the sensors is converted from the analog signal to the digital signal and the cancellation of the noise is done using these digitized values of the signal [14]. In general, the core assumption of active noise control is, that the recorded signal is additive. In other words, the signal is a sum of the noise signal and the desired signal, which bears all of the information [15]. The noise is assumed to not affect the desired signal. Thus can be written

$$y(t) = x(t) + n(t) \tag{2.1}$$

where $y(t)$ is the signal including both noise and the signal of interest as the function of time t, $n(t)$ is the noise signal and $x(t)$ is the signal of interest.

The task in active noise control is to find the noise signal $n(t)$ and separate it from the signal of interest $x(t)$ [16]. Most of the time, because the noise signal and the $x(t)$ are unknown, the endurable result is noise reduction from the original signal $y(t)$, not the complete cancellation. The reduction can also be done if the signal of interest $x(t)$ can be estimated, for example if the $x(t)$ is speech [15]. The cancellation is achieved by generating a sound wave of the opposite phase, which cancels the noise in question. This is called the principle of superposition [17].

2.2 Filters

In audio processing applications a filter is a key component [18]. The purpose of a filter in audio signal processing is to modify an audio signal towards an optimal target by removing unwanted features from it [19]. In noise control the filter suppresses the unwanted signal, in other words the noise, so the desired signal can be more distinguishable [20].

A filter can be either a physical device, which is referred to as an analog filter or an algorithm, which performs a mathematical operations on a signal digitally [21]. In this study, only the digital filters are discussed. Besides that, this study discusses linear filters, which means that the filter processes the frequency spectrum of the signal with a specific transfer function [22].

The aforementioned transfer function characterizes the filter by giving its frequency response. The transfer function of a linear time-invariant filter can be marked as H. Because in real-life situations the signal at time t is discrete, the Z-transform is made in order to convert the signal into frequency domain [9]. Therefore, the transfer function is marked as

$$H(z) = X(z)/Y(z) \tag{2.2}$$

where H is the transfer function, z marks a complex number and $Y(z)$ and $X(z)$ are the discrete Fourier transforms of the clean output signal $x(t)$ and noisy input signal $y(t)$ from the equation 2.1 [23]. Transfer functions of different filter designs vary, for example, the Butterworth filter has the transfer function based on the Laplace transform and the Wiener filter has its own transfer function based on minimizing the mean-squared error (MSE).

The aforementioned transfer function parameters are chosen to give the most adequate solution to the given filtering problem [24]. Besides the division between analogue and digital filters, filters can be classified as linear and non-linear filters [19].

2.2.1 Bandpass Filters

If there are known frequencies that do not contain any relevant information they can be filtered with high-pass, low-pass, or bandpass filters. For example, the most important factors for speech intelligibility can be found around 2000 Hz, where most of the consonants are heard and the most important range for speech is from 500 Hz to 4000 Hz [25]. It can be assumed that most of the relevant information lies in the aforementioned range when filtering an audio sample containing noise and speech. However, consonants like "s" have more energy in higher frequencies, therefore one can not tell the difference between "s" and "f" if the upper-frequency limit is for example 3500 Hz, which is a common upper-frequency limit in a mobile phone voice channel [26, 27]. Human sound spectre can be estimated to lay between 125 Hz and 8000 Hz, but some of the frequencies are less important to the intelligibility of the speech than others [25].

High-pass filter filters out certain frequencies below the given cut-off frequency, whereas a low-pass filter cuts off higher frequencies than the cut-off frequency and lets the low frequencies pass the filter. The bandpass filter is a combination of these two filters letting through certain a band of frequencies [28]. In digital audio signal processing there is a variety of IIR (Infinite Impulse Response) filters that are designed to do this kind of filtering, a few to mention elliptic filter, Butterworth and Chebyshev [29]. The three aforementioned filters are also known as the classical analog filters [30].

The general transfer function of the Butterworth filters is:

$$H(j\omega) = \frac{1}{\sqrt{1 + \frac{\omega}{\omega_c}^{2m}}} \tag{2.3}$$

where $H(j\omega)$ is the transfer function at angular frequency ω, m is the filter order and ω_c is the cut-off frequency, which is expressed as 2π times the cut-off frequency the calculations are made at [31].

Each of the IIR filters has a different length of the transition band, which is the range of frequencies between the passband and the stopband where the transition happens. They also have a different amounts of the ripple the filter generates. For example the elliptic filter has a short transition band but the ripple can be high whereas the Butterworth minimizes the ripple but the transition band is longer and gentler. With these filters both high-pass and low-pass filtering is possible, but they are not symmetric so in audio processing chaining high- and low-pass filters are more common than using a simple bandpass filter. [24]

In the figure 2.1 the difference between the elliptic and Butterworth filters are compared to each other. The horizontal axis is the normalized frequency, 1.0 being the Nyquist frequency. Nyquist frequency is one-half of the sampling rate of the processed signal. The frequency response consists of an amplitude response, vertical axis labeled *gain* in the 2.1 and a phase response, where the vertical axis is the *phase*. The gain can be expressed in decibels as

$$gain = 20 * log(\frac{V_{out}}{V_{in}}) \tag{2.4}$$

or with the general transfer function 2.3 for linear filter as

$$gain = 20 * log(H(j\omega)) \tag{2.5}$$

V_{out} being the magnitude of the output signal at given frequency and V_{in} being the magnitude of the input signal at given frequency [9]. For example, if the gain of the filter is $1/1000$ it is -60 Db. The transfer function however is more general and expressed in a complex form, whereas the gain is often calculated from a ratio between the input and the output [9]. In the low-pass filter's frequency response it can be seen that the elliptic filter has a very steep transition band, but in the picture the ripple effect is significant. Due to this all of the upper frequencies are not filtered. The Butterworth filters all of the upper frequencies, but the transition band has a much gentler slope.

The phase on the other hand describes the angular difference between two signals measured in degrees or radians [9]. For example for the signal that has 1 Hz frequency, the phase shift of 45°, the phase delay is 0.125 seconds. In the Figure 2.1 it can be seen that the phase response of the elliptic filter is nonlinear, while in the Butterworth filter it is nearly linear.

2.2.2 Adaptive Filter Theory

Adaptive filters are for the most part linear filters where the transfer function depends on the input. There are however some nonlinear adaptive filter applications, such as Volterra [32], but they are not discussed in this study. By adjusting the parameters according to the optimization algorithm, the filter can adapt to match the characteristics of the desired system and its features [20, 33]. Most often this means that the coefficients are adjustable by the adaptive algorithm [34]. In order to adapt without significant delay, adaptive filters need to be fast and not complex [35].

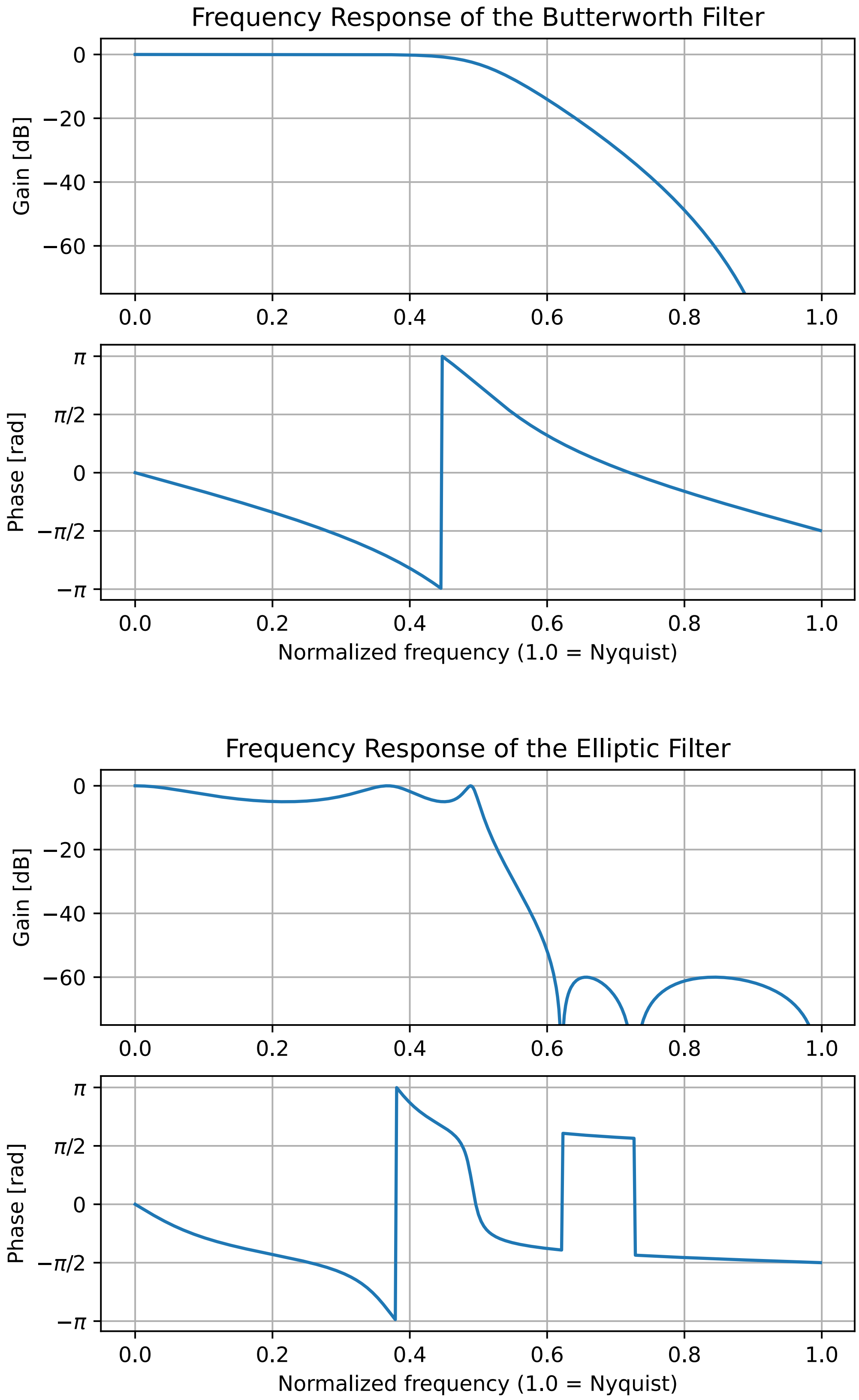

Figure 2.1. *Difference of the Butterworth and Elliptic Low-Pass Filters*

Unlike other traditional filters, adaptive filtering algorithms need no upfront knowledge of the signal, even though some adaptive filters can possibly benefit from the a priori knowledge [36]. However, all adaptive filters need historical knowledge of the latest filter state, while the filtering is done, in order to adapt. Consequently adaptive filters can handle both stationary and non-stationary signals [9]. Adaptive filtering algorithms aim to reduce the MSE (Mean-Square Error) towards an optimal value, which is called the statistical method of adaptive filtering [6].

Most of the time an adaptive filter is given a reference noise signal as an input, which it can use for better adaptation and noise canceling result [37]. In order to be useful the reference signal is assumed to be present in the noisy signal and both stationary and continuous. This way the noise in the noisy signal can be estimated with the help of the reference signal. Because the noiseless signal is achieved by subtracting the estimated noise from the noisy signal, the noise (which is present in the reference signal) should also be independent of the target sound [19].

The adaptive filter weights are chosen according to the reference signal and the estimated noise signal is calculated. First, the output of the adaptive FIR filter is calculated as

$$\hat{n}_1(s) = \sum_{k=0}^{M-1} w_k(s) n_2(s-k) \tag{2.6}$$

where $\hat{n}_1$ is the estimated noise, s is the sample being processed, w describes the filter weights and n_2 is the reference signal [16]. The term k is used for the k$^{\text{th}}$ element of the vector and M marks the order of the filter. When the estimated noise is subtracted from the unfiltered input signal $y(s)$, we get the estimated noiseless signal $\hat{x}(s)$ as an output, based on the equation 2.1, which is presented earlier

$$\hat{x}(s) = y(s) - \hat{n}_1(s) \tag{2.7}$$

and the error signal

$$e(s) = x(s) - \hat{x}(s) \tag{2.8}$$

Sometimes the reference signal is not available, for example, if there is only one microphone recording the noisy system or if the data used is prerecorded without the reference signal. In these situations, the reference signal can be calculated from the input signal by delaying the noisy process [38]. A simulated reference signal can be also made using just the error signal and its measurement [39]. After this, the generated reference signal is used in the processing in the same way as the real reference signal.

2.2.3 LMS Filters

The least mean square filter (*LMS filter*) is a linear adaptive filter, which was first introduced in 1960 by Widrow and Hoff [40]. LMS filter is designed to be computationally efficient and fast in comparison to other adaptive filters [41]. Due to these filter features the invention of the LMS filter has resulted in many different adaptations such as Normalized LMS (NLMS), Filtered-x LMS and Recursive LMS (RLS) [34]. Each of these has its own details and use cases, but the general idea of the LMS filter can be seen in the flowchart figure 2.2. In this research, the first-mentioned Normalized LMS filter adaptation is taken into closer inspection in the next subsection since it was used in the proposed solution.

The LMS filter is also gradient sample adaptive filter [9]. This means it is based on the descending gradient approach, an optimization algorithm, which tries to find a global minimum of a sequence. This is reached by taking small steps opposite the gradient and iteratively finding the global minimum. The filter h at time sample $s+1$ can be expressed

$$h_{s+1} = h_s - \frac{1}{2}\mu\nabla J[s] \tag{2.9}$$

where the step defined by a step-size $\frac{1}{2}\mu$ is taken opposite the cost function's gradient $\nabla J[s]$ from the current filter state h_s. The function of the gradient is

$$\nabla J(s) = -2p + 2Rw(s) \tag{2.10}$$

where R is the estimation of the autocorrelation matrix, p is the estimation of the cross correlation vector and w estimates the filter weights. In the LMS filter, the purpose is to modify these weights, in other words, the impulse response or the coefficients w, of the filter so that the error is as close to zero as possible. By updating the filter weights w the LMS filter tries to find this optimum filter weight

$$w_{s+1} = w_s - \mu e(s)n_2(s-i), \qquad \text{where } 0 < \mu < \lambda_{max} \tag{2.11}$$

where μ is the convergence rate of the filter, e is the error vector, n_2 is the reference signal, w_s marks the s^{th} element of filter weights or coefficients and λ_{max} is the maximum eigenvalue of the auto-correlation matrix [42, 43].

In the equation 2.11 the convergence rate μ, which is always a positive real number, defines the step-size of the gradient decent. Consequently, the filter stability and the learning rate are governed by the μ. Too large values of μ lead to an unstable filter as the filter is not able to adapt to the optimal solution, whereas too low values make the

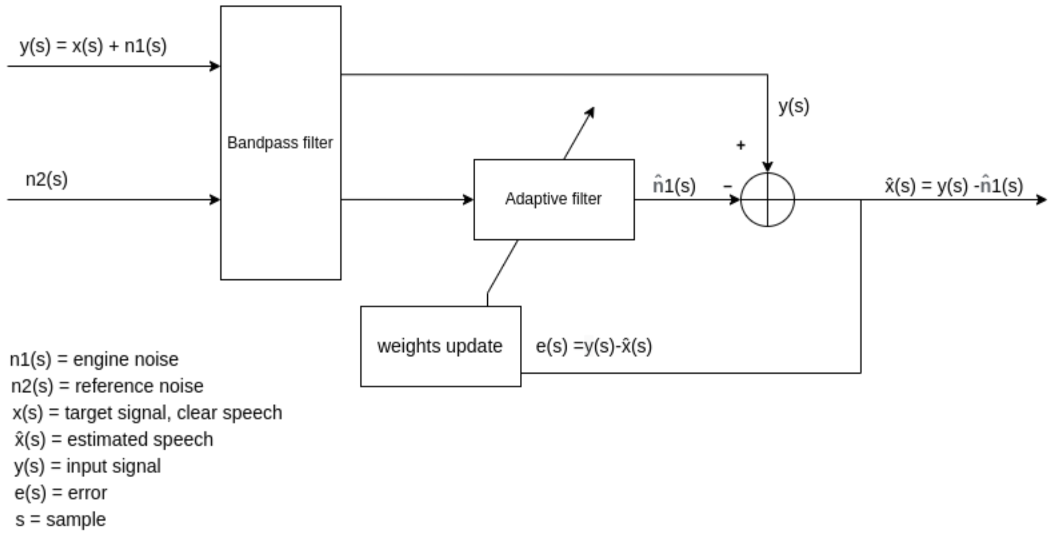

Figure 2.2. *Flow chart of the NLMS filter*

calculation slow. The optimal convergence rate μ can be chosen based on trial and error or former experience from the case [16]. It is noticeable that the adaptation is based only on the error at the current time [38].

Normalized LMS

Normalized LMS filter, in other words, the NLMS filter is modified from the LMS filter. It is invented to handle environmental changes better than the LMS algorithm because NLMS filter's convergence rate does not depend on the scale of the input [43]. As well as in the LMS filter, The stability, learning, and convergence rate of the NLMS filter are determined by the parameter μ [44]. However in the NLMS algorithm, the correction to the weights w_n is normalized and the calculation of the weights changes according to the energy of the input signal [10, 38]. This time-varying step-size μ, which is calculated as

$$\mu(s) = \frac{\beta}{|y(s)|^2}, \qquad \text{where } 0 < \beta < 2 \tag{2.12}$$

In the equation 2.12 y is the iteration s of the observed signal data vector and β is the normalized step-size [38].

In the literature, the NLMS filter was found to be better performing than the LMS filter. For example, Lampi [38], Rahman & Asadul Huq [10] and Tarrab & Feuer [43] stated that the converge rate and the Mean Squared Error of NLMS was found to be significantly better than LMS and other LMS based filters that were compared to the NLMS in the aforementioned studies. These studies suggest that using the Normalized LMS is currently the best practice when considering the use of LMS-based algorithms for sound enhancement purposes.

2.2.4 Wiener Filters

Wiener filters are optimal adaptive signal processing filters, that were first introduced in 1949 by Norbert Wiener, making a breakthrough invention in the statistical signal processing field [45]. They have been adapted in various filtering situations and modified to suit all kinds of tasks and are one of the most ubiquitous tools in signal processing [46]. According to Vaseghi et al. [9] Wiener formulation is also the basis of least square error adaptations such as adaptive LMS and NLMS filters.

The Wiener filter is based on frequency domain analysis and it is considered a block adaptive filter [9]. It tries to minimize the mean-square error, in other words, the difference between the desired signal and the estimated signal, which is the output of the filter. The filter takes in the noise-corrupted signal and the reference signal. These can be for example a recording from a human speaking near a moving car with a loud engine and a recording from the engine compartment. The reference signal is compared to the filter output in order to estimate the error and use it for the adapting [19].

In the optimal Wiener filter, the signal of interest is independent and uncorrelated from the noise signal. If the reference signal is not available, it can be also calculated from the input signal, for example using the local mean

$$mean = \frac{1}{N}\sum_{k=1}^{N} y_k \tag{2.13}$$

and the variance σ^2

$$\sigma^2 = \frac{1}{N}\sum_{k=1}^{N} y_k^2 - mean^2 \tag{2.14}$$

and using these aforementioned estimates the samplewise Wiener filter h can be constructed

$$h = mean + \frac{\sigma^2 - \upsilon^2}{\sigma^2}(y_{k\ldots N} - mean) \tag{2.15}$$

In these equations N is the length of the processing window, meaning the N-sized vector, from sample 1 to sample N at the first iteration, is taken from the noisefull audio signal y. This window is moved forwards at each iteration, so that at second iteration the vector is from sample $1+\mu$ to sample $N+\mu$, μ being the step-size [19]. In other words $y_{k\ldots N}$ is a vector of length N, k denoting the elements of the vector. υ^2 denotes the variance of the noise, which is estimated from the local variances. Python's Wiener algorithm found from the *scipy.signal* library uses this method for filtering [19]. The Wiener filter can not be used with non-stationary signals, because it needs apriori information on the statistics of the signal in order to adapt [47]. This leads to the core assumption that the original signal and the noise are stationary [48].

The transfer function of the Wiener filter can also be expressed in a different way with the power spectrum of the noise and the noise-free signal. This is known as the Wiener solution, $H(\omega)$, that is calculated with the equation

$$H(\omega) = \frac{P_x(\omega)}{P_x(\omega) + P_n(\omega)} \tag{2.16}$$

where $P_x(\omega)$ is power spectrum of the noise-free signal and $P_n(\omega)$ is power spectrum of the noise [49]. Because all of the elements are assumed to be real and non-negative, the Wiener filter affects only the signal's spectral magnitude, but not its phase. Notable also is that if there is no noise, the H becomes 1 and there is no need to filter, whereas if the noise is reaching infinity the H becomes 0.

The power spectra are assumed to be known before, but if they are not, they can be calculated from the samples of the signal y. Most spectral estimation techniques are based on the idea that the segment of this estimation, in other words, the correlation sequence can be found from the original signal. This way the exact correlation sequence $R(s)$ can be found when the following equation holds true

$$R(s) = \sum_{i=1}^{M} a_i e^{j\omega_i s} + z(s) = y(s) \tag{2.17}$$

where a_i is the power and ω the frequency location of the *i*th sinusoid and M denotes the number of sinusoids in the whole signal [19]. z is the zero-mean white noise and $y(s)$ is the noisy signal. The power spectrum $P(\omega)$ is obtained by taking the Fourier transform of $R(s)$. However, The Formula 2.17 holds true only for the ergodic random processes [19]

Usually, the Wiener filter is an FIR (Finite Impulse Response) filter because with discrete data the FIR filter is more straightforward to construct. However, IIR (Infinite Impulse Response) Wiener filter gives better results for the mean squared error than the FIR implementation [48]. In the IIR implementation, the impulse response of the time-domain is assumed to be infinite and the Wiener filter becomes more efficient because of that.

Due to the fact that the Wiener filter needs stationary noise for input, it is suitable for example in cases where the audio signal must be separated from stationary background noise. Even in cases where the noise signal is only short-time stationary the Wiener filter can be applied via short-time Fourier transform (STFT) in the time-frequency domain [46]. Using adaptive Wiener filter solution in the research [15] Abd El-Fattah discovered that it gave better denoising results than the traditional frequency-domain Wiener filtering, spectral subtraction, or wavelet denoising. The tests were executed with different noises and various evaluation methods for speech quality evaluation.

With the input signals where the signal-to-noise ratio (SNR) is high, meaning the desired signal has more power than the noise, the Wiener filtered signal may deteriorate [48]. However, in the cases where other filters usually decrease the intelligibility of the speech signal, the Wiener filter can produce a result signal where the quality is preserved [15]. The fact that the Wiener filter processes the signal in a continuous or discrete time domain is especially good for speech processing because speech is naturally varying in time domain [15]. This is also the motivation for choosing the Wiener filter to solve the stated problem in this Thesis.

Improving SNR in Wiener filtered signals the multichannel processing must be considered. When processing audio signals with multiple channels, there is also a spatial domain to be taken into account, not only the time-frequency domain, which is used in single-channel processing [50]. This is also the case with multichannel Wiener filtering. Due to the additional processing domain, the results of the noise reduction can be more advanced and preferable for the listener, unlike with single-channel solutions where the speech distortion may be significant [51, 52]. Theoretically, with multichannel processing the improvement in SNR can be achieved without the target speech signal being distorted, even when neither the reference signal nor any else apriori knowledge of the signal is accessible [52]. This is done by modifying Wiener filter to a suboptimal Wiener filter [52].

2.3 Processing Speech in Real-time Environment

When processing audio data in time-domain and real-time, a stream of short audio frames is fed to the processing as an input. In order to avoid discontinuation between frames, the filter state must be preserved between these audio frames [53]. For example, with a bandpass filter, this means the final filter delay values must be passed to the next frame to be processed. The discontinuities between the frames lead to a clicking sound in the processed audio [54]. Because the delay should be as short as possible, the noise reduction algorithm can not be mathematically complex with time-demanding calculations. In the case of adaptive filters, this means that the speed and the convergence rate of the algorithms are important and the accuracy of the results must be compared to the speed as well [20].

The most relevant aspects of speech signal noise reduction in a real-time environment are latency and speech intelligibility. The latency in signal processing expresses the time it takes for a data to travel from one point to another, for example how long it takes from a speech to travel from the person speaking to the audio recorder or to the listener's headphones. If the program meets the ITU standards about lip sync the delay should be less than 22 milliseconds [55]. For the case conversed in this thesis, meeting the aforementioned standard is not necessary, but it may possibly enhance the user experience. In the best case, speech enhancement makes speech more intelligible and increases the quality of the signal minimizing the distortion [56]. However, constructing the filter might also cause speech or noise distortion, which affects the listening experience [57]. Therefore on some occasions, the noise corrupted signal can be more pleasant to listen to than the enhanced signal even though the filter would enhance the desired signal [58].

The physical phenomenon called *speaking fundamental frequency* is a frequency, an average number of oscillations per second, that the structure of the speech signal has [59]. The pitch of the sound is a closely related term, being the perception of the fundamental frequency, in other words, the interpretation that people make from the heard signal [59]. An interesting phenomenon is, that a certain pitch can still be perceived even if the actual frequencies are not present in the signal [60]. The average speech pitch for an adult is about 100-300 Hz, depending on the speaker's age, a language they are speaking and living habits [61, 62]. Although the pitch is for example 120 Hz, after the signal is high-pass filtered with the cut-off frequency being 400 Hz meaning all frequencies lower than 400 Hz are non-existent, the perceived pitch is still 120Hz [59].

Speaking of high-pass filtering, the range of frequencies needed for the speech to be intelligible for non-tonal languages is 500-4000 Hz, so the intelligibility of the speech is preserved after the high-pass filtering with the cut-off frequency of 400 Hz [25, 63]. This range of speech intelligibility depends highly on the fact that the most important frequency range for consonants is between 2000 Hz and 4000 Hz [26]. The other factor affecting intelligibility is the signal-to-noise ratio, which is discussed more thoroughly in the next section 2.4.

2.4 Background for the Validation Method

Sound quality is difficult to be mathematically measured so that the results correlate with the subjective impression [64]. The way to substitute every aspect of human hearing is yet to be discovered and it possibly never will be [65]. The rate of convergence of the adaptive filter and the achieved mean-squared error can provide some clues about the quality of the sound [10]. Besides that, a huge variety of objective measures have been developed. In the further subsections, the signal-to-noise ratio and signal-to-distortion ratio are introduced.

Planning the listening tests may be found difficult since there is a huge variety of aspects in the test conditions that must be taken into account [66]. According to Karjalainen [33] the measurement of speech intelligibility and sound quality is based on three factors: the person who produces the speech, the medium through which the sound is passed, and the listener and their ability to receive and analyze the spoken message. Therefore the assessment of the sound quality is difficult. It depends on the factors that may not be reproducible in a reasonable manner. However, it is possible to have confidence in the results if the answers are approximately in the same area and not scattered on the validation scale [33]. Because the subjective results vary with different test participants, the grade scale should be as simple as possible. If the timbre of the voice is asked to be described the language of the questionnaire plays a role in the evaluation process [67].

2.4.1 Signal-to-Noise Ratio

The signal-to-noise ratio (SNR) is a simple validation method, which describes the relationship between the desired signal and the noise. It can also point out the effects the noise has on the desired signal. SNR is expressed in decibels so negative SNR values mean that the power of the noise is greater than the power of the signal. The SNR value of the signal is calculated with the following equation

$$SNR = 10\log_{10}\frac{P_x}{P_n} \tag{2.18}$$

where the P_x denotes the power of the desired signal and the P_n is the power of the noise signal [9]. From this also the formula using Root Mean Square (RMS), in other words, the voltage, can be derived, when the relationship

$$power = \frac{RMS^2}{resistance} \tag{2.19}$$

is used as

$$SNR = 10\log_{10}\frac{{RMS_x}^2/resistance_x}{{RMS_n}^2/resistance_n} = 10\log_{10}(\frac{RMS_x}{RMS_n})^2 = 20\log_{10}\frac{RMS_x}{RMS_n} \tag{2.20}$$

When calculating the SNR it must be remembered that the small properties of the desired signal and the noise affect the SNR. For example, if the amplitude of the signal spectrum is constant, the SNR can be altered by changing the bandwidth of the signal [68]. Therefore in some cases, the SNR value might be useless and distracting [33]. SNR also requires that the desired signal and noise powers can be estimated separately, which means that it does not work in cases where the powers can not be measured. However, when used properly the SNR can provide information about how much the desired signal is enhanced after the filtering process [68].

2.4.2 Source-to-Distortion Ratio

Source-to-distortion ratio (SDR) is a sound quality measurement that tells the performance of the overall separation between the sound source and distortion [69]. It is designed especially for the evaluation of blind source separation. The blind source separation in audio processing means that the method of how the different signals are mixed is not known beforehand. The SDR algorithm does the assumption that the true sources are known and some distortions that are allowed in the final signal are chosen. Some reference to the original signal must be given to the algorithm in order to calculate the SDR value. The sources are then decomposed based on orthogonal projections. The algorithm is as follows

$$SDR = 10 \log_{10} \frac{||x_{target}||^2}{||e_{interf} + e_{noise} + e_{artif}||^2} \tag{2.21}$$

where SDR is the source-to-distortion ratio, x_{target} is the true source signal and e_{interf}, e_{noise} and e_{artif} are the error estimates of the interference, noise and artifacts added to the signal. These values are different for each audio frame and therefore the perceived sound quality varies across time. [70]

SDR can be seen as SNR in practice, meaning the distortion is correlated with the clear speech signal, but noise is an unwanted signal independent from the speech signal [71]. This leads to the assumption that maximizing SDR leads to good sound quality. However, neither the SDR nor the other objective measurements correlate precisely with the subjective test results [33]. Therefore there are often several metrics used to describe the results, for example pythons function *fast_bss_eval* provides source-to-interference ratio (SIR) and source-to-artifact ratio (SAR) along with the SDR value [72]. The SIR describes the number of other sources heard in a source estimate, which is close to the sound leakage and the SAR tells the number of unwanted artifacts with relation to the true source.

2.4.3 Practices of Subjective Validation

Subjective validation refers to a situation where the validation is done by using listener's opinions of the produced sound. There are several standards guiding the validation procedure, for example, the standard "ITU-R BS. 1284-2" for the subjective assessment of sound quality [73], which is followed in this thesis when using the Comparative Mean Opinion Score (CMOS) seen in the figure 2.3. In the best-case scenario, the test is repeatable and gives the same results every time if the test conditions are the same. By increasing the number of listeners, the effect of an individual's opinion on the validation result is minimized. Despite of that, the test group should be chosen carefully. For exam-

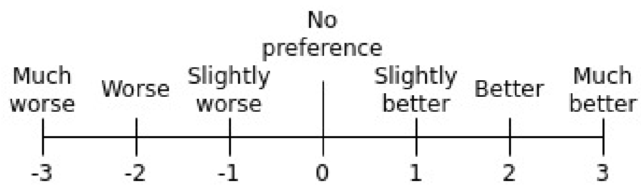

Figure 2.3. *Example of the CMOS*

ple, expert listeners who have participated in several listening tests tend to evaluate the sound differently than naive listeners who have less experience in evaluating sound [74].

One way of determining the reliability of the subjective test's results is to examine the distribution of the answers. If the results are scattered around the scale the results are usually not reliable. If, on the other hand, the results are focused around one parameter and there are only a few or no extreme values, the outcome of the inquiry can be considered trustworthy. [33]

When executing the hearing test, the environmental conditions should be as close to the final use situation as possible [33]. If the sound is meant to be listened to through a particular headphones they should be used in the test. The system meant to be used outdoors should be tested outdoors in the place close to the end-use environment. For example the wind, echo and the nearby traffic are considered when examining the results and developing the system further.

2.5 Other Methods Found in Literature / Previous Work

Besides the traditional signal processing methods used in this study, there is also a variety of other methods presenting a solution to the sound enhancement problem. One wide branch to take into account is machine learning algorithms and deep neural networks (DNN). Teaching the algorithm takes time, so when developing the system iteratively the machine learning method might not be the ideal approach [75]. However DNN does not necessarily need huge amounts of data to train the network, so the training can be quicker than other machine learning algorithms [76]. DNNs are widely used and they can be considered a good methods for speech enhancement as the methods presented in this theses work.

The hidden Markov model is another commonly used method in speech recognition, especially when the system needs not just to detect the speech but also to convert the speech to text. The hidden Markov model is a sequential and statistical method based on a chain of processes called the Markov process. A hidden Markov process describes a sequence of possible events, which have hidden states and the hidden Markov model tries to model the state transition probabilities of these hidden states. The model should be taught with for example speech data beforehand. [33]

One way of developing a sound-enhancing algorithm is to combine two or more methods. This is called the hybrid method or mixed methods. One example of this is to use improved Wiener filter and spectral weighting function together [77, 78]. This method is especially good for multichannel processing, which in audio processing means that the system has a mic-array [77].

Finally, a few of the earliest methods for speech enhancement must be mentioned, that is to say, the spectral subtraction and wavelet denoising used for enhancing speech intelligibility [15, 49]. Also besides the NLMS used in this thesis, the other LMS algorithms are common in literature in the area of sound enhancement, but as explained in the section 2.2.3 NLMS is the most suitable for this case [79, 80]. With LMS the dynamically adjusted filter length can be used, it may better the results because the optimal length of the filter can be found by the algorithm [81].

3. RESEARCH METHODOLOGY AND MATERIAL

This chapter describes the steps of the physical implementation of the soundscape-enhancing solution. All the filtering algorithms and their implementations are described along with the hardware, where the algorithms are meant to be executed. First, the overall working environment and its impact on the solution are covered. Next, the hardware is briefly discussed and the data is explained. After this, the actual audio processing algorithm is described starting from the preprocessing of the audio and followed by the description of the adaptive filters. The last chapter suggests other methods found in the literature, which were not covered in this research but could be implemented to solve this problem.

3.1 System Overview

The final solution consists of a development platform called Meluta MAX™. It is a standalone development platform for different real-time algorithms with various sensor connections. Concretely MAX™is built in a briefcase seen in figure 3.1. Several audio processing algorithms are implemented in this system, including the algorithms described in this thesis. Concerning the noise filtering algorithms, there is a switch in the User Interface (UI), where the user can choose whether or not the soundscape enhancement is on. There is also a second switch where the user can toggle between the two adaptive filter solutions. Regarding the final solution, the last-mentioned switch might be irrelevant since the end user is not assumed to understand the use of this switch. For the evaluation however it is important to be able to swap easily between the two adaptive algorithms.

In detail, MAX™has four microphones on top of it. Three of the microphones are placed in a triangular shape, they are called front (number 1), right (number 2) and left (number 3) microphones. The fourth microphone is placed at the center of the array higher than the three other microphones. This set up can be seen in the figure 3.1. In addition, a fifth microphone is placed in the vehicle's engine compartment to capture reference noise from the engine, seen in the figure 3.2. This microphone setup enables the system to be able to detect the direction where the target sound. The use of an accelerometer instead of microphones was considered because it was used successfully in literature, but it did not seem to give any additional benefit to the system [6]. Due to the microphone array,

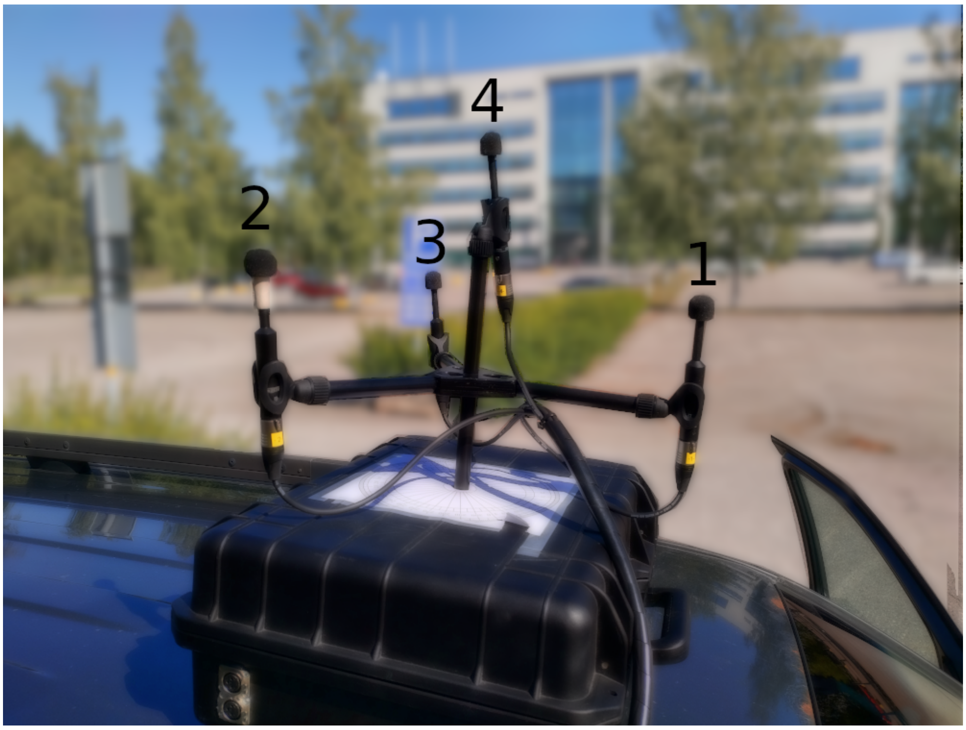

Figure 3.1. *Meluta MAX™platform's microphone placement in the microphone array*

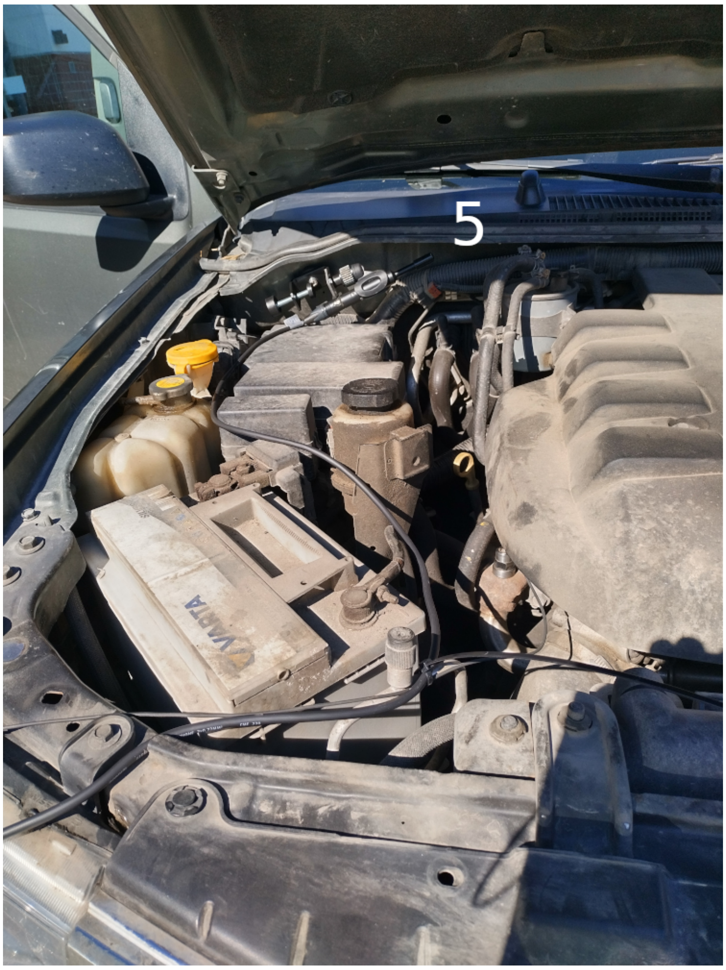

Figure 3.2. *Reference microphone placement in the engine compartment*

sound location detection is one of the capabilities of the MAX™, but the engine noise can complicate the detection process. This means the soundscape enhancement may also improve the accuracy of spatial detection.

The engine noise is most of the time periodic and therefore its characteristics are relatively easy to be found from the signal and filtered out [6]. Reassuring results for the success of adaptive filters for this kind of task came up in many research papers. For example, Thilagam et. al. managed to achieve 3-11 dB attenuation of environmental noise with an adaptive noise cancellation system [6]. Also Abd El-Fattah et. al. received encouraging results with their Wiener filter adaptation for speech enhancement [15]. This is one of the reasons why these algorithms were chosen for this case as well.

The code was done with Python version 3.8 with standard packages. Besides those packages *numpy* 1.21.2 and *scipy* 1.7.1 were used. The package *norbert* 0.2.1 was also tried with the Wiener filter algorithm, but since it did not compile with the overall program in the MAX™, the usage of this solution was abandoned.

The ultimate goal of the soundscape filtering was to make a human speech from outside the heavy equipment audible and perceivable to the machinist. The system needed to work in real-time so the machinist would react if they hear human sounds outside the heavy equipment they are operating. A few milliseconds delay was acceptable, but the algorithms were designed to be computationally efficient, so the sound going to the reference microphone would be filtered and reach the machinist as close to real-time as possible. Because of the computation efficiency, frequency domain processing was preferred over time-domain processing.

The program consists of three different blocks, as seen in Figure 2.2. First in the flow is a function that acts as an interface between the enhancing algorithms and the other algorithm implementations. This function feeds the sound blocks to the main function of the enhancement program. The engine noise was discovered to be low-frequency noise, adjusted between 90–250 Hz and depending highly on the crankshaft rotation and engine RPM [82]. The human hearing range is from 20 Hz to 20 000 Hz, so that is the first step of narrowing down the band [83]. Taken account the engine noise's range and the fact that in Chapter 2 the most valuable frequencies for speech intelligibility are stated to be between 500 Hz and 4000 Hz, and the target passband is set to be between 400 Hz and 4000 Hz. Thus the sound blocks are bandpass filtered with the bandpass filtering function, selecting only the frequencies containing human speech but minimal engine noise.

3.1.1 Bandpass filters

In the bandpass function, the signal's frequencies are filtered first with a high-pass filter and then with a low-pass filter. This is used instead of a bandpass filtering function since high-pass and low-pass filters are not symmetrical and therefore more accurate filtering can be achieved using them separately. The filter used in the final solution is a Butterworth filter. Also, elliptic filter was experimented with, but the ripple of the filter was too prominent. The *scipy.signal.butter* function was used with the order five (5) and cutoff frequencies mentioned earlier: 400 Hz for the high-pass and 4000 Hz for the low-pass filter. The filter's output is set to *sos*.

The infinite-impulse-response (IIR) filter design *sosfilt* is implemented because it is more stable and thus better for high-order filters like elliptic and Butterworth filters. Because of the real-time processing, the states of the high-pass and low-pass filters must be preserved. The *sosfilt* returns the filtered signals along with the filter states *zi*, which are passed to the next processing frame. In this project the *sosfilt* used was from the Pythons *scipy.signal* library.

3.1.2 Wiener filter Implementation

The results are returned to the main function, which feeds them to the adaptive filter selected by the user. The Wiener filter is directly from Python's *scipy.signal* library and it does not use the reference signal. Instead, the noise signal is estimated using non-parametric noise model, where the average of the local variance of the input signal are calculated with Equations 2.13 and 2.14 [84]. This calculation is done inside Python's function. Python's Wiener filter is a locally adaptive filter and it is designed for Gaussian noise, but despite of that the filter can adapt to other kinds of noises as well [19]. The processing is done in time domain.

Also, the multichannel processing was tested with the Wiener filter. However, in the early testing phase, the difference between single channel and multichannel enhancement results was found to be insignificant for the final purpose of the developed program and therefore the multichannel processing implementation was abandoned. The single channel Wiener filtering brings a few milliseconds delay, which is assumed to be tolerable. In multichannel processing, this delay could have been greater.

3.1.3 NLMS Implementation

The NLMS filter is designed and modified after the implementations found in the recent literature [38]. The filter convergence rate μ was experimented with different values between 0.1 to 0.001. Finally, the value 0.1 was decided as it gave the best result taking into account the speed of the calculations. The final order was decided to be 150. The order should be greater than the lag between the reference microphone and roof microphone, which was estimated to be 0.00583090379 seconds (about 94 samples) considering the speed of sound and few meter distance. The filtering is done with the equation

$$speech = y(s) - \hat{n}(s) \tag{3.1}$$

where y is the unfiltered signal and s indicates the processed frame (sample). The $\hat{n}$ is the estimated noise and it is calculated with the equation 2.6, where the reference vector is the engine noise. Both the weight vector w and the iteration of the reference vector are the same length as the filter order, which in this study is 150 with the step-size 1. This means the reference vector is windowed through by multiplying the M sized sample of the reference with the weight vector and the window is moved one step forward at each iteration [85]. The weight vector is adapted at every iteration i with the equation

$$w_{i+1} = w_i + \frac{\mu}{|{r_i}^T|^2 + 0.000001} e_i {r_i}^T \tag{3.2}$$

where the term 0.000001 is added to the equation to avoid the situation where $|{r_i}^T|^2$ is divided by zero, r_i being the reference signal at iteration i. It is so small, that it does not affect to the final result. ${r_i}^T$ is the transpose of r_i. The error vector e at iteration i is marked with e_i and it is calculated from the original signal as follows

$$e_i = x_i - \hat{n}_i \tag{3.3}$$

where the x_i is the estimated clean speech at iteration i and $\hat{n}_i$ is the estimation of the noise at iteration i.

3.1.4 Working Environment

The solution is specifically meant to be implemented in a human-controlled machine called Pantera™DP1500i, developed by Sandvik Mining and Construction oy. It is similar to the remote-controlled drilling equipment seen in the Figure 3.3, but with a cockpit. The machine is a top-hammer drill rig used in large quarries, open pit mines and construction sites [86]. In this kind of environment, there is usually high volume noise, which can be assumed stationary and used as a reference noise. In this thesis, only the engine noise is used as a reference noise, but in the future also the drill noise is meant to be filtered out from the soundscape.

The noise of the Pantera™DP1500i machine's engine is 60 dB louder than normal human speech, so the noise masks the speech completely. Besides that, the passive noise cancellation cancels out most of the surrounding voices as well. The system is focused on filtering out the human sounds from the environment and bringing them to the driver

Figure 3.3. *Sandvik remote controlled drilling equipment*

of the machine since human voices are the most critical ones from a safety point of view. This makes filtering the most time-critical part of the whole system and therefore the speed of the processing is more important than the high fidelity.

Also in the case that the machine is controlled from the control center and not from the cabin, it is impossible to separate human voices from the surrounding noise. The challenge of this kind of working environment can be the echoing of the noise. Even though the mine is an open pit mine, the rock reverberates the loud drilling noise. This demands further testing in the field, which were not possible during the research process for this thesis.

3.2 Data Collection

Many filtering solutions use desired signal (e.g. speech signal) as a reference for the construction of the adaptive filter. However wanting to process all of the environmental sounds and eliminate previously known engine sound, it is better to use the engine sound, in other words the noise source, as a reference for the filter construction. The noise source in this research is engine noise.

For simplicity, the noise was assumed to be stationary, meaning it is assumed to perceive specific parameters over an analysis period. In the test data, the reference noise came from a diesel engine, which was approximately 30 dB louder than human speech, which was estimated to be 50 dB [66]. In real life, the car diesel engine sound levels differed between 60 dB and 90 dB. In the field test in addition to the diesel engine, there was also a mining drill's engine, which can be estimated to be 40 dB louder than human speech. The drill on the machine produces a sound which is approximately 110 dB loud.

In order to develop the algorithm to work particularly on the engine noise filtering case, situational data was collected. There were two sessions of test data collection. All of the test data were stored in wav-files. The car used in the first two testing sessions was a Nissan Frontier pick-up truck with a diesel engine. The gear was not changed nor was the vehicle engine accelerated in the recordings, so the generated noise is constant and the engine noise can be assumed to be stationary. The MAX™modifies the sampling rate to 16 kHz for the sound enhancement algorithm, so the sampling rate of the recordings was irrelevant to the solution.

First test data recordings were executed with losslessly packed 24-bit data. The recorder was a Tascam recorder with four (4) channels. The set-up of the recorder can be seen in Figure 3.4. In addition to this, the one-channel microphone was placed in the engine compartment to be used as a reference noise, the placement is visualized in the figure 3.2. The diesel engine of the car acted as the noise source and while the recording was running, a test subject read a Finnish newspaper from a two-meter distance from the car. The recordings of the speech used in the development were one minute long.

The test subject was an average-height adult female. Both the test subject and the car were in the same positions during the whole recording session. The microphones were on the roof of the vehicle following the setup seen in the figure 3.1, where the first three microphones are in a triangular shape and the fourth is in the middle of the triangle about 20 cm higher than the other microphones. The speaker was standing on the left side of the vehicle, so microphone number 3 was nearest to the speaker. The recordings were done in the relatively quiet empty parking lot with tall buildings around it. The weather was sunny with low wind and about 15 degrees Celsius. It was noticed that the test subject raised their voice as the gear was shifted between the recordings and the car noise got louder. Therefore in the next session, the recordings were done with recorded speech played from a loudspeaker.

In the second test data recordings, the collected data was also losslessly packed 24-bit data. Because the examination of the direction of the sound was excluded from this study, only one (1) microphone was used to record the soundscape. Similar to the first data recordings, the reference noise was recorded from the engine compartment with one microphone. This recording session confirmed the system description delivered and approved by the customer. Like in the first recording session, a diesel engine car was

Figure 3.4. *Tascam, the recorder which was used to collect the test data*

used as a source of the reference noise.

As a target speech, there were recordings of the human speech, which were simulated to be at 5 m, 10 m, and 20 m distance from the car. Both male (lower pitch) and female (higher pitch) voices were present in the speech recordings. Besides that, an emergency vehicle siren recording was simulated similarly at 50-meter and 100-meter distances. The simulation was done by suppressing the audio according to

$$power = \frac{1}{distance^2} \tag{3.4}$$

so that if the distance is doubled, the power of the sound is 1/4 of the original sound. The baseline volume in the recording session was set using subjective estimation. The male voice sentence spelled out was *"Hey, stop right there"* and the female voice sentence was *"I'm not afraid, I was born for this"*. After the tests, the Wiener filter solution was found to be working, but the NLMS had a few bugs in the code.

Only one data collection was done in the real end-use environment. It was performed with the Pantera™DP1500i engine running and a driver inside listening to the speech coming from outside. The use of the loudspeaker system was difficult at the mining site, so real speech was used to test the system. The Pantera™DP1500i had so loud noise, that the speech was not audible at 10-meter and 20-meter distances, but at the 5-meter distance the driver inside the cabin told the speech was intelligible. The SNR with the drill on proved to be so low that this was foreseeable.

3.3 Other suggestions

In this thesis, the problem was approached with traditional signal processing methods. The machine learning algorithm which is taught to recognize sound sources could also be used to isolate human sounds from the noisy environments. This could be computationally more demanding for the system but is assumed to be as good as the solution presented in this thesis. With machine learning, the sound source localization can also be implemented in the same learning algorithm. [87]

A lot of things affect the sound quality. More testing could improve the results of the adaptive filtering, like changing the bit rate, and experimenting with different volumes and different speakers. Also using actual people instead of recordings could bring out the flaws of the system as well as can the usage of heavy equipment instead of the pick-up truck.

4. RESULTS AND ANALYSIS

To validate the applicability of the algorithms described in this thesis, both subjective and objective measurements were used. The first section of this chapter describes the evaluation process introducing the signal-to-noise ratio and subjective evaluation. The used methods are rationalized and explained. The next chapter summarizes and speculates the results received from the objective evaluation, followed by the section describing the results of the objective evaluation method. The last section presents methods that were not used in this thesis but might improve the results in future research and considers how the used evaluation methods could be improved.

4.1 Evaluation Procedure

Taken account that the enhancement was made for an end-user and not just the research purposes in mind, a subjective evaluation needed to be done. There is also no objective method that would give exactly correlated results with the subjective hearing. Subjective tests were executed in an environment mimicking the end-users working environment and vehicle's engine noise. The data for objective evaluation was recorded in the subjective tests.

There is a wide variety of objective methods for estimating speech intelligibility and filter performance, some are designed for a specific case and some are more universally applicable. Even though these objective measurements do not exactly correlate with subjective testing, they give a guideline of how the algorithm works. Also, they are easier to execute and the conditions do not affect the result as much as with listening tests. Some of the objective speech evaluation algorithms require a reference signal in other words a clean speech signal to make the evaluation, while other algorithms are non-intrusive. [88]

The objective evaluation however gives a guideline of how the sound enhancement system works, especially in the case where the subjective tests have only a few test subjects. This is the reason why the objective evaluation was done in this research. Since the recorded audio was used in the testing phase, intrusive speech evaluation algorithms can be used. Since the thesis converses on speech enhancement, the chosen objective measurement was the source-to-distortion ratio along with the signal-to-noise ratio.

4.2 Objective Evaluation

For the assessment of the objective validation, the signal-to-noise ratio and the signal-to-distortion ratio were used. The results give a guideline about how well the NLMS and Wiener filters performed the noise reduction. Also, the convergence rate of the mean-square error was examined.

The calculation of SNR was done according to the formula 2.18. For example in the 5m distance when a human is speaking in a normal voice, their sound pressure level is about 45 dB and when shouting, 63 dB [26]. The sound pressure level of the car engine used in the listening tests was measured to be 80 dB. For the Panthera drilling machine, the same measurement produced a value of 125 dB. This makes the SNR without the filter about -35 dB with the car engine and -80 dB with the heavy equipment.

The SNR of the final results was calculated over the short root-mean-squared (RMS) values of the signal with the function 2.20 where the RMS was estimated by using the equation

$$RMS = \sqrt{\frac{1}{M}\sum_{k=1}^{M} x_k^2} \tag{4.1}$$

where M is the length of the vector and x_k indexes the element of the noise or speech vector. The estimation was done by taking a frame of a size 2048 samples from the signal, the sampling frequency being 16 kHz, and calculating the RMS with that. Over 10-second long signal the mean of all RMS values was calculated and the final RMS value was received for the signal. The signal used for the filtered signal RMS was the recording with the car engine and filtered speech, whereas the RMS of the noise signal was the recording from the same location with the car engine on but no speech.

In this way, the SNR before processing was about -33.2 dB, which correlates with the estimated SNR mentioned previously. For the processed audio the SNR was about -8.0 dB. This means the noise still overpowers the signal, but the filter managed to improve the relation by 25.2 dB. The values were similar for both of the filters, Wiener filter receiving only 0.2 decimals higher improvement for SNR value than the NLMS filter.

For the SDR the Python function *bss_eval_sources* from the package *fast_bss_eval* was used [72]. The test was executed for two different audios, the first was recorded at the testing site with microphone placements describer in 3.1 and 3.2 and the second was simulated audio. The simulation was done by adding the recorded engine noise to the clear speech signal and filtering the resulting noisy audio. The reference engine noise used for NLMS was a different recording than the one added to the signal, but since the recordings of the reference signal were from the same microphone as the added noise,

Audio	Absolute SDR	Difference to unfiltered
Unfiltered	-16.47	-
Wiener	-15.53	0.94
NLMS	-15.48	0.99
Unfiltered simulated	-6.29	-
Wiener simulated	-6.17	0.12
NLMS simulated	3.86	10.15

Table 4.1. *Test results for SDR calculation*

the filtering result was better than in real life situation.

The speech signals of the recorded audios seen in 4.1 were aligned with each other. The lags in the Bluetooth connection of the MAX platform could have affected the results so that the speech signals did not align perfectly with the non-simulated signals. This impact was estimated to be no more than one decibel. Also in the field recordings, the engine noise was relatively silent so it could have resulted to lower SDR values since the speech was clearly audible in both unfiltered and filtered cases. In the calculations, the clear speech signal was used as the benchmark and other signals were compared to that.

From the table 4.1 the results of the SDR calculation can be seen. For the unfiltered signal, the SDR was -16.47 dB and for the Wiener filtered signal the received value was -15.53 dB and for the NLMS filtered signal -15.48 dB. The difference between the unfiltered and Wiener filtered signal was 0.94 dB and with the NLMS 0.99 dB. This difference is nearly insignificant but can be explained by the silent engine noise.

For the simulated audios the results for SDR differentiated more. The SDR value for the NLMS algorithm was relatively good, 3.86 while in the literature the best absolute value for SDR in the field of music separation is 7.68 [89]. The difference between the SDR value of the unfiltered and NLMS filtered simulated audio was high, 10.15, which can be due to the reference noise and the mixed noise being close to each other. For the Wiener filter, the simulated audio provided an insignificant difference between unfiltered and filtered audio. Subjectively there was a difference between the engine noise levels, but it did not enhance the speech so that it would have had an effect on the SDR.

At the calculation of MSE the time-frequency units, in other words the frames, are treated as individuals so that the calculations of one frame do not affect the other frames. MSE is also a linear scale, while human hearing is logarithmic. Even though this leads to the fact that MSE does not match the human auditory perception, the convergence of the MSE value can be used to evaluate the performance of the algorithm. The convergence of the MSE means that over the processing of the whole audio, the MSE should convergence towards some constant number and, in the figure 4.1 this value can be estimated to be

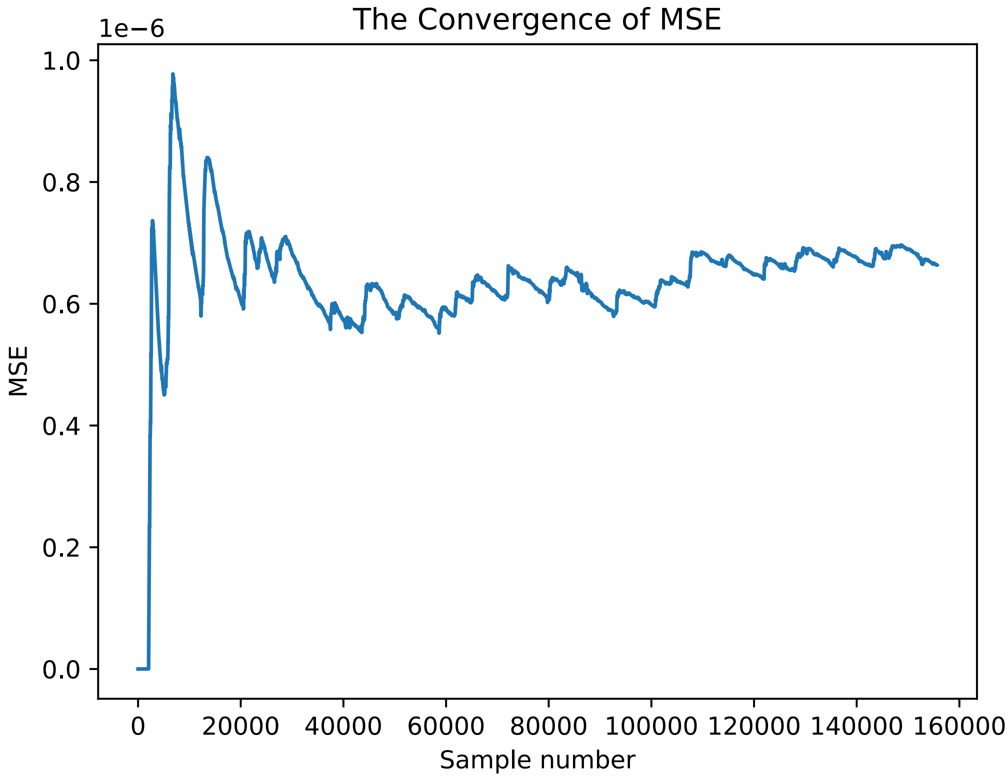

Figure 4.1. *Convergence of MSE in one processing frame*

$0.65 * 10^{-6}$ in this study. If the MSE would not converge the algorithm would not be adapting.

The afore-mentioned values indicate that the algorithms achieved moderate noise reduction. Further mathematical evaluation could be done in order to get more precise values describing the behavior of the two filtering solutions. However, in this study, the subjective evaluation is more significant and it is discussed in the following section.

4.3 Subjective Evaluation

The test setup was done with one active roof mic on top of the Nissan Pathfinder, a large diesel engine car, and one microphone as a reference mic in the engine compartment. The subjective evaluation was done using the CMOS described in 2.4.3. A loudspeaker was placed about five (5) meters from the passenger side door and a male voice saying the phrase *"Hey, stop right there"* was played from the system. The speech from 10 meters and 20 meters distance were simulated and played from the system as well. The engine noise differed between 60 dB and 90 dB, while the speech at 5-meter distance was set to 60 dB. According to the measurements in Wenmaekers' and Hak's research [90] the experienced level of the speech at 20-meter distance was about 30 dB in this study, which was recognized to be nearly inaudible.

Test nr	Much worse	Worse	Slightly worse	No preference	Slightly Better	Better	Much Better
T1							
T2							
T3							
T4							
T5							
T6							
T7							
T8							
T9							
T10							

Figure 4.2. *Form used in objective validation tests*

Both male and female voices were considered as the played sample and both were tested by the programmer. In the evaluation procedure, the male voice was used. This is because the male voice has been proven to give better results in subjective evaluation than the female voice. Like Larsby states, the female voice has been discovered to require a greater signal-to-noise ratio to receive the same evaluation score as the male voice [91].

4.3.1 The Listening Test Setup

There were both expert listeners and naive listeners in the test subject group, about a 50/50 ratio. The expert listeners had participated in several listening tests before or worked closely with audio enhancement tasks. In the test setup, the listener sat on the passenger's seat and the test coordinator sat on the driver's side and controlled the test by playing the audio and explaining the test procedure along the way.

At first the listeners familiarized themselves with the test materials, the test procedure, and the environment the test was executed in. The form filled in the listening tests can be seen in the figure 4.2. The listeners were given Bluetooth headphones that had a passive noise control feature. The control over the on/off switch of the filter was given to the listeners, so they had time to internalize the task in question and switch the filtering when they felt comfortable to. One task lasted about 15 seconds and the whole testing procedure was over in 10 minutes, which meets the ITU criteria about subjective sound evaluation [73].

There were two arranged listening tests. The first was done in March, at the empty parking filled with snow. The temperature was about minus ten degrees Celsius. The wind was low and there was no other traffic near the testing site. In the first tests, all of the six test subjects were part of the company in which this thesis was done, so there might have been a bias. In this testing round the reference microphone was unavailable, so only the Wiener filter was tested. Also contrary to the second test round the emergency vehicle sirens were tested.

The second test were executed in June at the University parking lot. The wind was decent but had an influence on the sound quality, which is prominent from the test results. There were also few by-passers walking or cycling past the testing site, so it could have caused a minor disturbance. The emergency vehicle sirens were left out from this test round because the target of interest in this study is enhancing the speech signal. In addition, the duration of the tests would have been too long with the siren sounds. With listening tests lasting over 15 minutes, the ITU standard [73] recommends a break for the listeners, so the length of the test does not affect the results and the tests were wanted to be below that time limit.

4.3.2 Test results

As seen in the table 4.2 the first tests provided really good results. All of the inputs reached the expected output or were even better than expected. The bias may be improved the results, but more likely the expected results were reached due to the fact that all of the participants were really aware of what they should evaluate and what was the "better" or "much better" filtering results supposed to sound like. In this round, all of the participants also spoke the same native language as the instructor, Finnish. This rules out the misunderstanding due to the language barrier in the test situation.

In the case where the simulated speaker was 20 meters from the car, the phrase was still audible to the listeners during the first tests. In the second tests, many participants commented that due to the wind and the speaker volume the phrase was inaudible and the evaluation of the situation was difficult. In the real-life case a human speaking 20 meters apart from the listener can be inaudible also at a near-noiseless environment, so the test results according to this case might be worse because of the overall volume, not the goodness of the system.

At every test, the human speech at 5-meter distance got the best results of the enhancement speech. This can be caused by the fact that the speech volume made the speech more audible despite the noise in the background. The louder speech volume might have made the abating of the noise more outstanding as well. With the simulated ambulance siren, the system filtered also the traffic noise from the played recording, so the good performance with the siren can be contingently explained with that.

In the second tests, the native language of most of the participants was other than Finnish or English. This can have an effect on the results by decreasing the scoring of the filtering result because the understanding of the assignment might have been harder. Even though all of the participants understood English, the phrase could have been evaluated as less intelligible, because it was not in the native language of the listener [92].

In the second tests, the cases where only the environment sounds were present with and

Input	Expected output	Actual output	Acceptance criteria	Test status
Human speaking at 5 m distance, engine is running	>2	3	>1	Output better than expected
Human speaking at 10 m distance, engine is running	>2	2	>1	Expected output received
Human speaking at 20 m distance, engine is running	>2	2	>1	Expected output received
Simulated ambulance siren at 50 m distance, engine is running	>2	2	>1	Expected output received
Simulated ambulance siren at 100 m distance, engine is running	>2	2	>1	Expected output received
Silent environment, engine is not running	>0	0	>0	Expected output received
Silent environment, engine is running at operating RPM	>0	2	>0	Output better than expected

Table 4.2. *Test results for first tests*

Input	Mean	Median	Mode
Human speaking at 5 m distance, engine is running	1.2	2	2
Human speaking at 10 m distance, engine is running	0.7	1	-
Human speaking at 20 m distance, engine is running	-0.1	0	-
Silent environment, engine is not running	1.8	2	2
Silent environment, engine is running at operating RPM	1.6	2	2

Table 4.3. *Test results for Second tests, Wiener filter*

Input	Mean	Median	Mode
Human speaking at 5 m distance, engine is running	0.6	1	-
Human speaking at 10 m distance, engine is running	0.6	0	0
Human speaking at 20 m distance, engine is running	0.1	0	-
Silent environment, engine is not running	1.6	1	1
Silent environment, engine is running at operating RPM	1.2	1	-

Table 4.4. *Test results for Second tests, NLMS filter*

without the engine noise were evaluated much better (2) with the Wiener filter and better (1) with the NLMS filter. This can be caused by several reasons. One is that due to the bandpass filter some frequencies of the speech were filtered and the speech quality got subjectively worse scores due to that. For example, consonants in high frequencies such as *s* were distorted. However the intelligibility did not suffer from the bandpass filtering, but the test subjects could have evaluated the quality of the speech rather than the intelligibility or the suppressing of the engine noise. With this realization, the focus of the test subject may have been on the speech and how it changed due to the filtering and without the speech, the focus was on the abating of the engine noise which lead to better results. The filtering worked also by suppressing the wind sounds and without the speech, this phenomenon was noticed by the listeners.

The filter where the Wiener algorithm was implemented performed better than the NLMS filter. The NLMS filter took more time to develop and the result might not have been optimal. The NLMS filter might have altered the speech more, so the speech was more intelligible with the filtering made by the Wiener filter and the surroundings might have sounded more pleasant to the listener.

It might be irrelevant to consider the error margin in this test because the group of test listeners was so small. However, the tables 4.3 and 4.4 show that in some cases the mode could not be calculated because the answers were so scattered around the scale. Especially in the cases where the median is zero, the answers were widely spread throughout the scale and the result might not be reliable.

4.4 Future Work

At first, the use of the accelerometer was considered. It might have been useful in filtering the engine noise because in theory the vibration of the engine is picked by the accelerometer, but the surrounding sounds such as the birds singing or wind are not. This way all of the surrounding sounds are brought to the listener and only the engine noise is filtered. The downside of the proposed method is that the filter concentrates on one type of noise and it does not affect the wind or passing traffic noise, which the filters in this thesis filtered also. Even though the accelerometer was left out of this thesis, it can be advantageous to research it in the future.

In the perfect testing setup, the tests would be performed with a Panthera series machine so the noise source is as authentic as possible and the speech enhancement is optimal for the target environment. The NLMS algorithm could be developed further so that it filters the wind noise and focuses on the speech only. For example, if the convergence of the NLMS algorithm is followed, it could be turned off or switched to the Wiener filter when there is no reasonable convergence rate.

Regarding the listening tests, the number of test listeners could have been greater. This minimizes the error margin and makes the final evaluation of the subjective tests easier. Also, one error-generating case is the language. Both the language of the test phrase played by the speakers and the native language of the test listeners can affect the results. In addition, one thing which might need improvement is how clearly the assignment can be given to the test listeners. In the tests described in this thesis, the directions were given sometimes in Finnish and sometimes in English depending on the listener's native language. If this is done in a more consistent manner, using only one language for prepping, the results will not be corrupted by the language spoken in the test setup. Also, the test phrase could be in the same language as the listener's main language so the intelligibility is optimal.

In the future of MAX™, the customer's comments about the performance of the system could be considered. Those are the two-way communication between driver and person outside the vehicle and the speakers distance detector. Detecting how far the speaker is from the vehicle and figuring out the direction of the sound source could improve the safety aspects of the heavy equipment.

Related to the work in the project, the multichannel processing could be combined with the location detection algorithms. That is one huge field of signal enhancement left out from this research. The MAX™briefcase has four microphone arrays, so the multichannel processing is possible to be executed with the same platform as the single channel processing done in this thesis. Multichannel processing enables location-based enhancement. In real life, people can focus their attention on one sound direction and improve the audibility of the speech in this way [66]. When considering the multidimensional technique for speech enhancement, the effects of the context in the subjective evaluation must be taken into account [87]. When enhancing the surrounding sounds of the vehicle and bringing them to the headphones, there are only two output channels for the multi-channel microphone system.

5. CONCLUSION

In this thesis, two adaptive filters, the Wiener filter, and the normalised LMS filter were applied to filter the engine noise and thus enhance speech in heavy equipment soundscape. The programmed filtering system consisted of a bandpass filter targeting the frequencies of human speech and the previously mentioned adaptive filter. The system was implemented on Meluta MAX platform as a part of a bigger project including sound direction detecting and radars.

The system was tested multiple times and developed further according to the test results. The completed program was tested and validated in a real-time environment to investigate the achieved results. In the final listening tests, the system was found to be working moderately and they managed to better the original audio. The main challenges were the other noises than the engine noise and setting the floor value for intelligibility in the listening tests to achieve convergent answers. The mathematical evaluation proved to be difficult to do, but the results correlated with the subjective evaluation. This indicates that the results can be considered reliable and they were as good as anticipated.

It was noticed that there is still room for improvement and further development of the Wiener filter and NLMS filter for the enhancement of speech. Also, there were several aspects in the subjective evaluation, that could have been organized better so that the error would be minimal. The design of the listening tests lead to a situation where the results were not as good as anticipated. Regardless the system enhanced the speech instead of deteriorating it. The most recognizable results from the subjective tests are, that at the first test the wiener-filtered human speech at 5-meter distance was evaluated *much better* (3) and at the second test the wiener-filtered silent environment was evaluated *better* (2). It is also worthy of mentioning that none of the filtering cases worsened the speech or the environment sounds.

Most of the errors were caused by the delay of the processing and the convergence capability of the algorithm. The NLMS produced discontinuities because of the processing speed of the Python operations used in the calculation of the NLMS. The Wiener filter had no such issue, but since the filter was taken directly from the Python library, it was not specially designed to do speech processing or engine noise filtering. Consequently, the system could be improved by designing the Wiener filter to target the engine noise.

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